Bank Customer Churn: Predictive Modeling and Risk Scoring

Unified Mentor | Data Analyst Internship | Dataset: European_Bank.csv

Abstract and research objective

This project analyzes 10,000 banking customer records to predict churn (Exited=1). Observed churn is 20.4%. Four supervised classification models were evaluated using a stratified 80/20 train-test split. The best model by held-out ROC-AUC is Random Forest, with ROC-AUC 0.865. Scores are predictive estimates, not verified future outcomes.

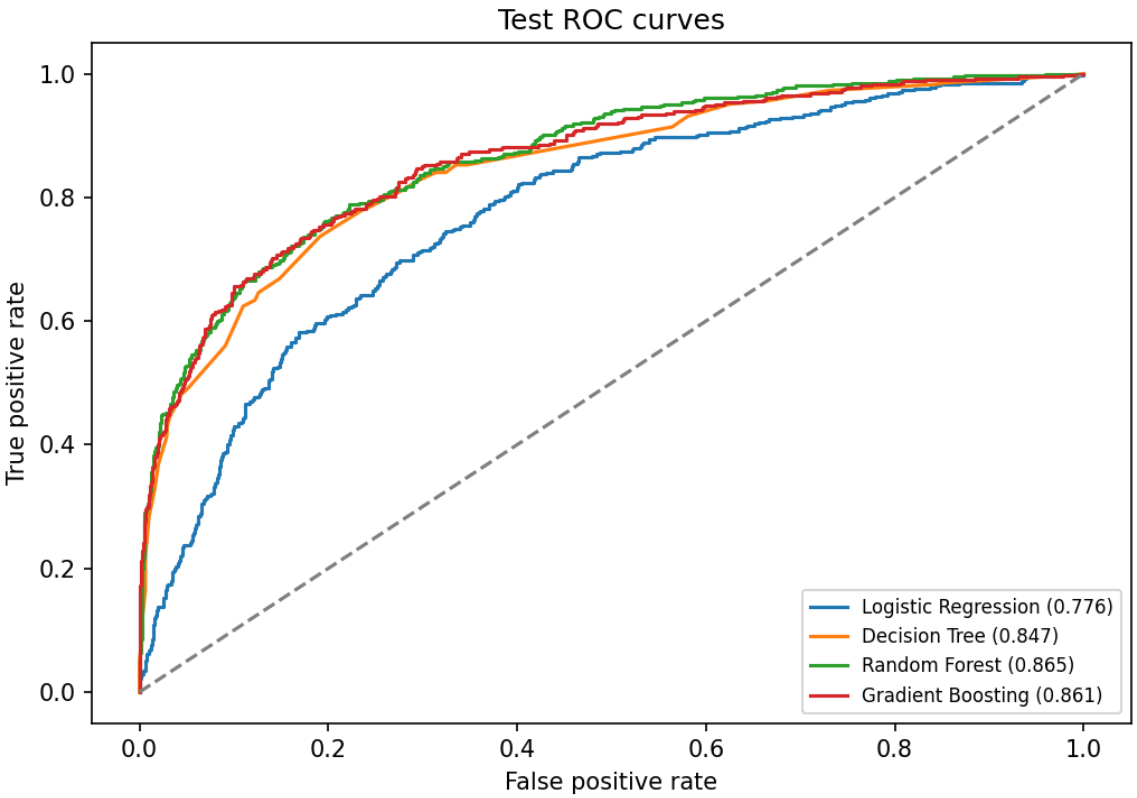
Methodology

CustomerId and Surname were excluded as identifiers. Year was excluded because all rows share the same year (2025). Geography and Gender were one-hot encoded; numerical columns were imputed and standardized. Derived variables include balance-to-salary ratio, engagement-products interaction, age-tenure interaction, and products-per-tenure. Preprocessing is fitted only on training data via scikit-learn pipelines.

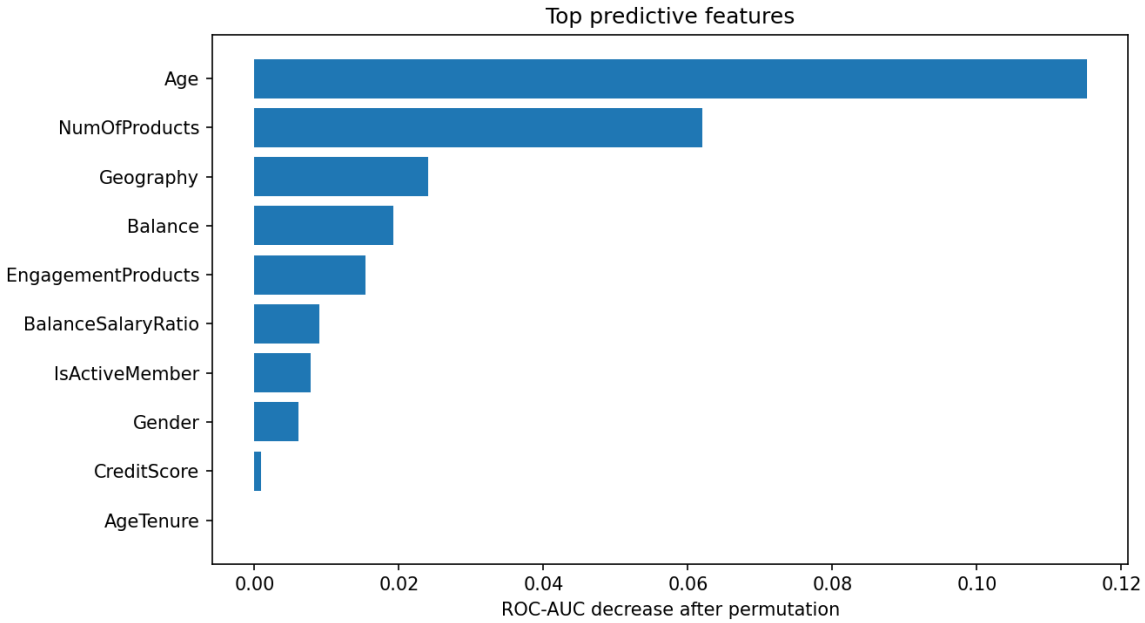
Model evaluation on 2,000 held-out customers

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.849	0.628	0.627	0.627	0.865
Gradient Boosting	0.860	0.734	0.489	0.587	0.861
Decision Tree	0.861	0.788	0.430	0.556	0.847
Logistic Regression	0.710	0.383	0.698	0.494	0.776

Risk decision threshold 0.44 was chosen using a validation subset of training data, not the test set. Test confusion matrix (true rows: retained, churned; predicted columns: retained, churned): [[1378, 215], [127, 280]]. Test F1 at this threshold: 0.621.


Test ROC curves
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (0.776)
Decision Tree (0.847)
Random Forest (0.865)
Gradient Boosting (0.861)

Feature importance and interpretation



Top predictors by held-out permutation importance: Age, NumOfProducts, Geography, Balance, EngagementProducts, BalanceSalaryRatio. Importance indicates model reliance, not a causal explanation. SHAP and partial dependence are suggested future extensions, not implemented in this package.

Recommendations for stakeholders

Prioritize outreach to high-risk customers, validate contact costs and retention uplift in controlled campaigns, and avoid using risk scores as a basis for adverse banking decisions. Monitor performance and calibration on newer customer cohorts. Review model fairness across demographic groups, privacy, consent, and applicable banking regulations before any real deployment.

Limitations

The dataset is a static labeled snapshot and cannot establish causality or real-world retention impact. No external validation or prospective deployment was performed. Risk bands (Low below 0.30; Medium 0.30 to below 0.60; High at least 0.60) are illustrative operational groupings, separate from the validated binary threshold. Results should not be represented as live bank performance.

Reproducibility

Run python train.py to reproduce model training and outputs, then streamlit run app.py for the dashboard. Fixed random seeds and a stratified split are used. Python dependencies are listed in requirements.txt.